JEE Mathematics: Quick Review with Grading and Practice Map - Indefinite Integration (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (58 questions; a question that appears in two of these is counted once): High = 5 or more questions, Medium = 1 to 4 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

Define the terms: integral, primitive, integrand, anti-derivative.

Model Answer

If $\frac{d}{dx}\phi(x) = f(x)$ then $\int f(x)\,dx = \phi(x) + C$: f is the integrand, ϕ is a primitive (anti-derivative) of f, C is the constant of integration (so the integral is called indefinite).

Two primitives of the same f differ by a constant; a given value such as $f(a) = k$ fixes C.

Ex: $\frac{d}{dx}x^x = x^x(1 + \ln x) \Rightarrow \int x^x(1 + \ln x)\,dx = x^x + C$.

Grade - High

Practice map, main concept: Sheet: J-I 1; J-M 34 | DPP: 2(9); D5(6)

Also used in: Sheet: J-I 5; J-II 85; C-II 25 | Question Bank: 1, 5, 7, 10, 12-13, 19

Standard Integrals

Question 02

Write all five sets of standard integral results.

Model Answer

(1) $\int x^n dx = \frac{x^{n+1}}{n+1} + C\ (n \neq -1)$, $\int \frac{dx}{x} = \ln|x| + C$

(2) $\int e^x dx = e^x + C$, $\int a^x dx = \frac{a^x}{\ln a} + C$

(3) $\int \sin x\,dx = -\cos x + C$, $\int \cos x\,dx = \sin x + C$

(4) $\int \sec^2 x\,dx = \tan x$, $\int \operatorname{cosec}^2 x\,dx = -\cot x$, $\int \sec x \tan x\,dx = \sec x$, $\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x$

(5) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a}$, $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$, $\int \frac{dx}{x\sqrt{x^2 - 1}} = \sec^{-1}|x|$, $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|$

With $ax + b$ in place of x divide the result by a. First simplify the integrand ($e^{\ln f} = f$, identities, product to sum).

Ex: $\int \cos 2x \cos 3x\,dx = \frac{1}{2}\int(\cos 5x + \cos x)\,dx = \frac{\sin 5x}{10} + \frac{\sin x}{2} + C$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2, 14; J-II 1-3, 20, 99, 108; J-M 80; C-II 27 | DPP: 1(1-8, 10-12, 14, 16-27); 2(1-2, 6, 8, 10-17); 3(27); 5(7); D1(13, 15); D2(14, 17-18)

Also used in: Sheet: J-I 16, 35; J-II 8, 21, 42, 56, 58, 78, 97; J-M 56-57; C-II 2, 15 | DPP: 2(9); 4(13, 18)

Question 03

How to integrate linear/linear?

Model Answer

Write the numerator as k(denominator)+ constant, so $\dfrac{ax+b}{cx+d} = \dfrac{a}{c} + \dfrac{m}{cx+d}$; then integrate term by term.

Ex: $\displaystyle\int \frac{x}{x+3}dx = \int \left(1 - \frac{3}{x+3}\right) dx = x - 3\ln|x+3| + C.$

Grade - **Low**

Practice map, main concept: DPP: 1(13, 15)

Also used in: DPP: 3(20)

Question 04

How to integrate $\sin^n x$, $\cos^n x$ (small n)?

Model Answer

Lower the power by multiple-angle formulas: $\sin^2 x = \dfrac{1-\cos 2x}{2}$, $\cos^2 x = \dfrac{1+\cos 2x}{2}$, $\sin^3 x = \dfrac{3\sin x - \sin 3x}{4}$, $\cos^3 x = \dfrac{3\cos x + \cos 3x}{4}$; $\sin^4 x = (\sin^2 x)^2$.

Ex: $\displaystyle\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$; $\displaystyle\int \cos^3 x\,dx = \frac{3\sin x}{4} + \frac{\sin 3x}{12} + C.$

Grade - **Medium**

Practice map, main concept: none

Also used in: Sheet: J-M 19 | DPP: 4(9)

Question 05

Integrate $\dfrac{1}{1+\sin x}$ (and $\dfrac{1}{1+\cos x}$).

Model Answer

Multiply by the conjugate: $\dfrac{1}{1+\sin x} = \dfrac{1-\sin x}{\cos^2 x} = \sec^2 x - \sec x\tan x$; or use half angles: $1+\cos x = 2\cos^2\dfrac{x}{2}$.

$$\int \frac{dx}{1+\sin x} = \tan x - \sec x + C, \qquad \int \frac{dx}{1+\cos x} = \tan\frac{x}{2} + C$$

Grade - **Low**

Practice map, main concept: DPP: 2(7); D3(18)

Also used in: Sheet: C-II 7 | DPP: 4(8); 5(5); D2(11)

Question 06

How to integrate 1/Quadratic?

Model Answer

Complete the square: $D>0 \to \displaystyle\int \frac{dx}{X^2-a^2}$, $D=0 \to \int \frac{dx}{(ax+b)^2}$, $D<0 \to \int \frac{dx}{X^2+a^2}$.

Ex: $\displaystyle\int \frac{dx}{x^2-6x+10} = \int \frac{dx}{(x-3)^2+1} = \tan^{-1}(x-3) + C.$

Grade - **Medium**

Practice map, main concept: Sheet: J-II 43, 51 | DPP: 2(5); D2(1)

Also used in: Sheet: J-I 4; J-II 27, 38, 68, 98; J-M 70 | DPP: 3(17-18, 21); D1(7, 14); D3(20)

Integration by Substitution

Question 07

What is the general method of substitution, and which forms does it give directly?

Model Answer

$$\int f(\phi(x))\,\phi'(x)\,dx \xrightarrow{\phi(x)=t} \int f(t)\,dt$$

Split the integrand so that one factor is the derivative of the expression inside the other. Direct forms: $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$, $\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + C$.

Ex: $\int \frac{x^5}{1+x^6}dx$, put $1+x^6 = t$: $= \frac{1}{6}\ln(1+x^6) + C$.

Grade - High

Practice map, main concept: Sheet: J-I 4-5, 9, 15-16, 25; J-II 5-8, 10, 21-22, 46, 50, 58, 67, 85, 97-98; J-M 17, 60, 79; J-A 1, 6, 13, 16; C-I 1, 7, 24; C-II 1-2, 11-12, 15 | DPP: 1(9); 2(18-20, 23); 3(1-2, 5-7, 9-11, 13-15, 18-24, 28-29); 5(15); D1(5, 7-8, 17-18, 22) | Question Bank: 1, 5, 14, 19

Also used in: Sheet: J-I 6-7, 10, 23, 28, 85; J-II 14, 16, 36, 38, 108-109; J-M 3; J-A 32, 41; C-II 6-8, 22, 26, 37 | DPP: 4(17); 5(16-17); D1(3, 9); D2(18, 21); D5(14) | Question Bank: 10-12, 15-16

Question 08

Write the results of integration of $\tan x$, $\cot x$, $\sec x$ and $\operatorname{cosec} x$.

Model Answer

$$\int \tan x\,dx = \ln|\sec x| + C, \qquad \int \cot x\,dx = \ln|\sin x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C = \ln\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + C$$

$$\int \operatorname{cosec} x\,dx = \ln|\operatorname{cosec} x - \cot x| + C = \ln\left|\tan\frac{x}{2}\right| + C$$

Grade - Medium

Practice map, main concept: Sheet: J-M 57

Also used in: Sheet: J-I 14; J-II 6, 20; J-M 35; J-A 6; C-II 3 | DPP: 2(16); 3(2) | Question Bank: 18

Question 09

Integrate $\frac{1}{a\sin x + b\cos x}$.

Model Answer

Write $a\sin x + b\cos x = r\sin(x+\alpha)$, $r = \sqrt{a^2+b^2}$, $\tan\alpha = \frac{b}{a}$; then $\int = \frac{1}{r}\int \operatorname{cosec}(x+\alpha)\,dx = \frac{1}{r}\ln\left|\tan\frac{x+\alpha}{2}\right| + C$.

Ex: $\cos x + \sqrt{3}\sin x = 2\sin\left(x + \frac{\pi}{6}\right)$, so $\int \frac{dx}{\cos x + \sqrt{3}\sin x} = \frac{1}{2}\ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{12}\right)\right| + C$.

Grade - Low

Practice map, main concept: Sheet: J-I 3

Also used in: Sheet: J-II 26 | DPP: 5(5)

Question 10

What kind of substitution is used to deal with an inverse trigonometric function?

Model Answer

If the derivative of the inverse function is present, put the inverse function $= t$ ($\sin^{-1} x = t$, $\tan^{-1} x = t$). If $\sin^{-1}$, $\tan^{-1}$ acts on an algebraic expression, put $x =$ a trigonometric function so that it simplifies (e.g. $x = a\tan^2\theta$ in $\sin^{-1}\sqrt{\frac{x}{a+x}}$).

Ex: $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}}dx = \frac{(\sin^{-1} x)^2}{2} + C$; $\int \frac{e^{\tan^{-1} x}}{1+x^2}dx = e^{\tan^{-1} x} + C$.

Grade - **Medium**

Practice map, main concept: Sheet: J-II 11-12; C-II 17-18 | DPP: 2(24); 3(4, 8, 12, 16); 4(5); D3(4) | Question Bank: 9

Also used in: DPP: 2(6); 4(2, 14)

Question 11

What kind of substitution is used to deal with a logarithmic function?

Model Answer

When $\frac{1}{x}$ is present with a function of $\ln x$, put $\ln x = t$, $\frac{dx}{x} = dt$. (If $\frac{1}{x}$ is absent, use parts with $\ln x$ as first function.)

Ex: $\int \frac{dx}{x\ln x} = \ln|\ln x| + C$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 8; J-II 4; J-A 41 | DPP: 2(22); 3(17); 5(10); D2(13, 22); D5(14)

Also used in: Sheet: J-I 11; J-II 6, 20 | DPP: 4(7, 15-16); D1(6)

Integration by Trigonometric Substitution

Question 12

Which trigonometric substitution should be used?

Model Answer

$a^2 + x^2$: $x = a\tan\theta$; $a^2 - x^2$: $x = a\sin\theta$; $x^2 - a^2$: $x = a\sec\theta$; $\sqrt{\frac{a-x}{a+x}}$: $x = a\cos 2\theta$; $\sqrt{\frac{x}{a-x}}$: $x = a\sin^2\theta$.

Ex: $\int \frac{dx}{(1+x^2)^2}$, $x = \tan\theta$: $\int \cos^2\theta\, d\theta = \frac{1}{2}\tan^{-1} x + \frac{x}{2(1+x^2)} + C$.

Grade - **Medium**

Practice map, main concept: Sheet: J-M 19, 35; C-II 8 | DPP: D2(4, 11, 21); D5(10) | Question Bank: 15

Also used in: Sheet: J-II 13, 32; C-II 4, 11, 23 | DPP: 2(21); 3(31); D1(20)

Question 13

Write the results of $\int \frac{dx}{\sqrt{a^2-x^2}}$, $\int \frac{dx}{\sqrt{x^2+a^2}}$ and $\int \frac{dx}{\sqrt{x^2-a^2}}$.

Model Answer

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left|x+\sqrt{x^2+a^2}\right| + C, \qquad \int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left|x+\sqrt{x^2-a^2}\right| + C$$

Grade - **Low**

Practice map, main concept: Sheet: C-II 6 | DPP: 2(4); 4(13)

Also used in: Sheet: J-II 33, 50, 57 | DPP: 3(16, 30)

Question 14

How to integrate $\dfrac{1}{\sqrt{\text{Quad.}}}$?

Model Answer

Complete the square. Leading coefficient > 0: $\int \dfrac{dx}{\sqrt{X^2 \pm a^2}}$ (or $\int \dfrac{dx}{|ax+b|}$ if $D = 0$); leading coefficient < 0: $\int \dfrac{dx}{\sqrt{a^2 - X^2}}$.

Ex: $\int \dfrac{dx}{\sqrt{x^2+2x+5}} = \ln\left|x+1+\sqrt{x^2+2x+5}\right| + C.$

Grade - **Medium**

Practice map, main concept: Sheet: J-A 8 | DPP: 4(12); D1(11)

Also used in: none

Integration by Parts

Question 15

Write the formula for integration by parts.

Model Answer

$$\int \mathrm{I} \cdot \mathrm{II}\, dx = \mathrm{I} \int \mathrm{II}\, dx - \int \left(\frac{d}{dx}\mathrm{I} \int \mathrm{II}\, dx \right) dx$$

Ex: $\int x \sin x\, dx = -x\cos x + \sin x + C.$

Grade - **High**

Practice map, main concept: Sheet: J-I 7; J-II 13-14; J-A 2, 5, 21; C-I 23; C-II 3 | DPP: 4(1, 6, 8-9); D1(2, 24-25); D2(7, 15, 19) | Question Bank: 4

Also used in: Sheet: J-I 8-9; J-II 11-12, 19, 40, 42, 67; J-A 16; C-II 16-18 | DPP: 4(2, 5); D1(13); D3(4)

Question 16

How would we choose the first and the second function?

Model Answer

ILATE: the first function is the one that comes first in Inverse, Logarithmic, Algebraic, Trigonometric, Exponential; the second function must be easy to integrate.

Ex: $\int x \ln x\, dx = \dfrac{x^2}{2}\ln x - \dfrac{x^2}{4} + C$ ($\ln x$ first).

Grade - **Low**

Practice map, main concept: Sheet: J-I 6 | DPP: 4(4)

Also used in: DPP: 4(1); D1(2, 24)

Question 17

What is the trick of integration by parts and in which type of functions is it used?

Model Answer

(i) A single function whose integral is not known ($\ln x$, $\sin^{-1} x$, $\tan^{-1} x$): take 1 as the second function. (ii) Cyclic integrals ($e^{ax}\sin bx$, $\sec^3 x$): the same integral comes back; move it to the left side. (iii) Forcing: write the integrand as $\mathrm{I} \cdot \mathrm{II}$ with a convenient factor.

Ex: $\int \tan^{-1} x\, dx = x\tan^{-1} x - \dfrac{1}{2}\ln(1+x^2) + C.$

Grade - **High**

Practice map, main concept: Sheet: J-I 84-85; J-II 40, 78, 82-84; J-M 3, 58, 69; C-I 2; C-II 13, 23 | DPP: 4(2-3); D1(4); D2(8) | Question Bank: 12, 18

Also used in: Sheet: J-II 4; J-M 17; C-II 5 | Question Bank: 4

Question 18
Write the results of $\int \sqrt{a^2 - x^2}\,dx$, $\int \sqrt{x^2 - a^2}\,dx$ and $\int \sqrt{x^2 + a^2}\,dx$.

Model Answer

$$\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$$

$$\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln\left|x + \sqrt{x^2 - a^2}\right| + C$$

$$\int \sqrt{x^2 + a^2}\,dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln\left|x + \sqrt{x^2 + a^2}\right| + C$$

Grade - Medium

Practice map, main concept: Sheet: J-II 57 | DPP: 3(30)
Also used in: Sheet: J-II 17 | Question Bank: 19

Question 19
Write the results of $\int e^{ax}\sin bx\,dx$ and $\int e^{ax}\cos bx\,dx$, and prove them.

Model Answer

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\sin bx - b\cos bx) + C$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\cos bx + b\sin bx) + C$$

Proof: integrate by parts twice (cyclic); I reappears with factor $-\frac{b^2}{a^2}$; solve for I. Ex: $\int e^{2x}\sin 3x\,dx = \frac{e^{2x}}{13}(2\sin 3x - 3\cos 3x) + C$.

Grade - Medium

Practice map, main concept: Sheet: J-M 7; C-II 22 | DPP: 4(7, 10)
Also used in: none

Integration of Rational Function

Question 20
How to integrate 1/linear?

Model Answer

$$\int \frac{dx}{ax + b} = \frac{1}{a}\ln|ax + b| + C$$

Ex: $\int \frac{dx}{3x + 2} = \frac{1}{3}\ln|3x + 2| + C$.

Grade - Low

Practice map, main concept: none
Also used in: none

Question 21

How to integrate polynomial/linear?

Model Answer

Divide: $\dfrac{P(x)}{ax+b} = Q(x) + \dfrac{r}{ax+b}$ (remainder $r = P(-b/a)$).

Ex: $\displaystyle\int \frac{x^3+x^2+x+4}{x+1}dx = \int \left(x^2+1+\frac{3}{x+1}\right)dx = \frac{x^3}{3} + x + 3\ln|x+1| + C.$

Grade - Medium

Practice map, main concept: DPP: 2(3)

Also used in: DPP: 3(23) | Question Bank: 5

Question 22

How to integrate $\sqrt{\text{Quadratic}}$?

Model Answer

Complete the square and use the three results of 18: $\sqrt{X^2-a^2}$, $\sqrt{X^2+a^2}$, $\sqrt{a^2-X^2}$ ($D=0$: $\sqrt{(ax+b)^2} = |ax+b|$).

Ex: $\displaystyle\int \sqrt{x^2+2x+5}\,dx = \frac{x+1}{2}\sqrt{x^2+2x+5} + 2\ln\left|x+1+\sqrt{x^2+2x+5}\right| + C.$

Grade - Medium

Practice map, main concept: Sheet: J-II 17

Also used in: Question Bank: 17

Question 23

How to integrate linear/quadratic, linear/$\sqrt{\text{quadratic}}$ and linear$\cdot\sqrt{\text{quadratic}}$?

Model Answer

Write linear $= A\dfrac{d}{dx}(Q) + B$ (compare coefficients). The A-part is $\displaystyle\int \frac{Q'}{Q}$, $\displaystyle\int \frac{Q'}{\sqrt{Q}} = 2\sqrt{Q}$ or $\displaystyle\int Q'\sqrt{Q} = \frac{2}{3}Q^{3/2}$; the B-part is item 6, 14 or 22. Polynomial/quadratic: divide first.

Ex: $4x+3 = 2(2x-2)+7$, so $\displaystyle\int \frac{4x+3}{\sqrt{x^2-2x+2}}dx = 4\sqrt{x^2-2x+2} + 7\ln\left|x-1+\sqrt{x^2-2x+2}\right| + C.$

Grade - Medium

Practice map, main concept: Sheet: J-I 35; J-II 15, 38, 56; C-II 4 | DPP: 3(31); 4(11) | Question Bank: 17

Also used in: Sheet: C-II 14 | DPP: D1(12, 19)

Two Classic Integrals

Question 24

Write both results of the two classic integrals and prove them.

Model Answer

$$\int e^x[f(x)+f'(x)]\,dx = e^x f(x) + C, \qquad \int [xf'(x)+f(x)]\,dx = xf(x) + C$$

Proof: $\dfrac{d}{dx}[e^x f(x)] = e^x(f+f')$ and $\dfrac{d}{dx}[xf(x)] = f + xf'$ (or parts on $\int e^x f\,dx$). Ex: $\int e^x(\sin x + \cos x)\,dx = e^x \sin x + C$.

Grade - Medium

Practice map, main concept: Sheet: J-I 12; J-II 18, 42, 47; J-M 36; C-I 9; C-II 40 | DPP: 4(22); D1(6, 23); D2(9)

Also used in: Sheet: J-II 82; C-I 23 | DPP: D5(1)

Question 25

When would we apply the classic integral results?

Model Answer

When the integrand is $e^x \times$(something) and the something splits into a function plus its derivative (simplify, split fractions, use $\frac{x-1}{x^2} = \frac{1}{x} - \frac{1}{x^2}$ etc.); similarly $f + xf'$ without e^x.

Ex: $\int e^x \dfrac{x-1}{x^2} dx = \int e^x \left(\dfrac{1}{x} - \dfrac{1}{x^2}\right) dx = \dfrac{e^x}{x} + C.$

Grade - Medium

Practice map, main concept: Sheet: J-I 10-11 | DPP: 4(14-17, 19-21, 23) | Question Bank: 2

Also used in: Sheet: J-II 18, 47; J-M 36

Question 26

What to do in questions of classic integral if an expression other than x is in the power of e?

Model Answer

Put the exponent $= t$ (e.g. $kx = t$, $\ln x = t$), or use $\int e^{kx}[kf(x) + f'(x)]\, dx = e^{kx} f(x) + C$.

Ex: $\int e^{2x}(2\tan x + \sec^2 x)\, dx = e^{2x} \tan x + C.$

Grade - Low

Practice map, main concept: DPP: 4(18); D5(1)

Also used in: none

Question 27

Write the result of $\int e^{g(x)} \left(f(x)g'(x) + f'(x)\right) dx$.

Model Answer

$$\int e^{g(x)} \left(f(x)g'(x) + f'(x)\right) dx = e^{g(x)} f(x) + C$$

Reason: it is $\dfrac{d}{dx}\left[e^{g(x)} f(x)\right]$. Ex: $\int e^{x^2} \left(2x \ln x + \dfrac{1}{x}\right) dx = e^{x^2} \ln x + C.$

Grade - Medium

Practice map, main concept: Sheet: J-I 13, 34; J-II 19; J-M 46; J-A 25; C-I 3, 8; C-II 37 | DPP: D1(3); D2(5) | Question Bank: 13

Also used in: Sheet: J-M 34; C-I 2; C-II 12 | DPP: D1(23); D2(8)

Integration of Trigonometric Functions

Question 28

How to integrate $\sin^m x \cos^n x$ if at least one of m, n is odd?

Model Answer

m odd: put $\cos x = t$; n odd: put $\sin x = t$ (keep one odd power for dt, convert the rest with $\sin^2 x + \cos^2 x = 1$).

Ex: $\int \sin^5 x \cos^4 x\, dx = -\dfrac{\cos^5 x}{5} + \dfrac{2\cos^7 x}{7} - \dfrac{\cos^9 x}{9} + C.$

Grade - Low

Practice map, main concept: DPP: 3(3); 5(3)

Also used in: DPP: D2(7)

Question 29

How to integrate $\sin^m x \cos^n x$ if both m and n are even?

Model Answer

Convert higher powers into multiple angles: $\sin x \cos x = \frac{1}{2}\sin 2x$, $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$.

Ex: $\int \sin^2 x \cos^2 x \, dx = \dfrac{x}{8} - \dfrac{\sin 4x}{32} + C.$

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-II 8 | DPP: D2(4); D5(10)

Question 30

How to integrate $\sin^m x \cos^n x$ if $m+n$ is a negative even integer?

Model Answer

Write it as $\tan^m x \sec^{-(m+n)} x$, keep $\sec^2 x$ for dt and put $\tan x = t$.

Ex: $\int (\sin x)^{1/3} (\cos x)^{-7/3} dx = \int \tan^{1/3} x \sec^2 x \, dx = \dfrac{3}{4}\tan^{4/3} x + C.$

Grade - Medium

Practice map, main concept: Sheet: J-I 23; J-II 9, 24-25; J-A 32; C-II 9 | DPP: 3(26); D1(9) | Question Bank: 11

Also used in: Sheet: J-M 13, 21

Question 31

How to integrate $\dfrac{1}{a\sin^2 x + b\cos^2 x + c\sin x \cos x + d}$?

Model Answer

Multiply numerator and denominator by $\sec^2 x$ ($d = d\sec^2 x \cos^2 x$, i.e. $d \to d(1+\tan^2 x)$) and put $\tan x = t$: it becomes $\int \dfrac{dt}{\text{quadratic in } t}$.

Ex: $\int \dfrac{dx}{\sin^2 x + 4\cos^2 x} = \int \dfrac{dt}{t^2+4} = \dfrac{1}{2}\tan^{-1}\left(\dfrac{\tan x}{2}\right) + C.$

Grade - Medium

Practice map, main concept: Sheet: J-II 23, 61, 68; J-M 37; J-A 31; C-II 33 | DPP: 5(1, 4, 6) | Question Bank: 6-7

Also used in: Sheet: J-II 41; C-II 23

Question 32

How to integrate $\dfrac{1}{a\sin x + b\cos x + c}$?

Model Answer

Write $\sin x = \dfrac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}$, $\cos x = \dfrac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$ and put $\tan\dfrac{x}{2} = t$.

Ex: $\int \dfrac{dx}{1+\sin x + \cos x} = \ln\left|1+\tan\dfrac{x}{2}\right| + C.$

Grade - Medium

Practice map, main concept: Sheet: J-II 59; J-M 70 | DPP: 5(2, 5); D1(14, 16); D3(20)

Also used in: none

Question 33

How to integrate $\dfrac{l\sin x + m\cos x + n}{a\sin x + b\cos x + c}$?

Model Answer

Write $\text{Nr} = A(\text{Dr}) + B\dfrac{d}{dx}(\text{Dr}) + C$ and compare coefficients: $\int = Ax + B\ln|\text{Dr}| + C\int \dfrac{dx}{\text{Dr}}$ (last part by 32).

Ex: $\sin x = \frac{1}{2}(\sin x + \cos x) - \frac{1}{2}(\cos x - \sin x)$, so $\displaystyle\int \frac{\sin x}{\sin x + \cos x}dx = \frac{x}{2} - \frac{1}{2}\ln|\sin x + \cos x| + C$.

Grade - Medium

Practice map, main concept: Sheet: J-I 17; J-II 52; J-M 20 | DPP: 5(8) | Question Bank: 8

Also used in: none

Question 34

How to integrate when $(\sin x + \cos x)$ or $(\sin x - \cos x)$ is in the numerator and $\sin 2x$ in the denominator?

Model Answer

$\text{Nr} = \sin x + \cos x$: put $\sin x - \cos x = t$, $\sin 2x = 1 - t^2$. $\text{Nr} = \sin x - \cos x$: put $\sin x + \cos x = t$, $\sin 2x = t^2 - 1$. ($\sin x$ or $\cos x$ alone: split as half-sum / half-difference.)

Ex: $\displaystyle\int \frac{\sin x + \cos x}{8 + \sin 2x}dx = \int \frac{dt}{9 - t^2} = \frac{1}{6}\ln\left|\frac{3 + \sin x - \cos x}{3 - \sin x + \cos x}\right| + C$.

Grade - Low

Practice map, main concept: Sheet: J-II 27-29; C-I 15; C-II 10 | DPP: 5(9, 17); D2(6)

Also used in: DPP: 1(9)

Question 35

How to integrate when $(\sin x + \cos x)$ or $(\sin x - \cos x)$ is in the denominator and $\sin 2x$ in the numerator?

Model Answer

Write $\sin 2x = (\sin x + \cos x)^2 - 1$ or $\sin 2x = 1 - (\sin x - \cos x)^2$, divide, and use 9 for $\displaystyle\int \frac{dx}{\sin x \pm \cos x}$.

Ex: $\displaystyle\int \frac{\sin 2x}{\sin x + \cos x}dx = \sin x - \cos x - \frac{1}{\sqrt{2}}\ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{8}\right)\right| + C$.

Grade - Low

Practice map, main concept: Sheet: J-II 26

Also used in: none

Integration by Partial Fraction

Question 36

When would we use the partial fraction method?

Model Answer

For a rational function $\dfrac{P(x)}{Q(x)}$ whose denominator factorises. If $\deg P \geq \deg Q$ divide first $(T(x) + \dfrac{P_1}{Q})$.

A substitution ($x^2 = t$, $e^x = t$, $\sin x = t$) often makes the integrand rational first.

Ex: $\displaystyle\int \frac{x\,dx}{(x^2+2)(x^2+1)}$, $x^2 + 1 = t$: $= \dfrac{1}{2}\ln\dfrac{x^2+1}{x^2+2} + C$.

Grade - Medium

Practice map, main concept: Sheet: J-II 16 | DPP: 5(12); D2(10); D3(28)

Also used in: Sheet: J-I 25; J-M 28, 54; C-II 10 | DPP: 1(19); 3(22)

Question 37

How to break the fraction if the denominator is a product of non-repeating linear factors?

Model Answer

$$\frac{P(x)}{(x-a_1)(x-a_2)\cdots(x-a_n)} = \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \cdots + \frac{A_n}{x-a_n}$$

A_k: cover $(x-a_k)$ and put $x = a_k$. Ex: $\int \frac{dx}{(x-1)(x-2)(x-3)} = \frac{1}{2}\ln|x-1| - \ln|x-2| + \frac{1}{2}\ln|x-3| + C$.

Grade - Medium

Practice map, main concept: Sheet: J-M 1, 28, 54

Also used in: DPP: 5(6, 10, 15); D1(8)

Question 38

How to break the fraction if the denominator is a product of linear factors, some of them repeating?

Model Answer

$$\frac{1}{(x-a)^k(x-b)} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k} + \frac{B}{x-b}$$

Find A_k, B by putting $x = a$, $x = b$; the rest by comparing coefficients. Ex: $\frac{1}{x^2(x+1)} = -\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x+1}$, $\int = \ln\left|\frac{x+1}{x}\right| - \frac{1}{x} + C$.

Grade - Low

Practice map, main concept: Sheet: J-II 30-31; C-II 7, 20 | DPP: 5(11)

Also used in: Sheet: J-II 32, 34

Question 39

How to break the fraction if the denominator is a product of linear and quadratic factors?

Model Answer

$$\frac{1}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

An irreducible quadratic factor gets a linear numerator. Ex: $\int \frac{dx}{(x-1)(x^2+1)} = \frac{1}{2}\ln|x-1| - \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\tan^{-1}x + C$.

Grade - Medium

Practice map, main concept: Sheet: J-II 32; J-M 21, 38; C-II 25 | DPP: 3(25)

Also used in: Sheet: J-II 24 | DPP: 5(12); D1(17); D2(10); D3(28)

Miscellaneous

Question 40

How to apply the 'kutur putur' profile?

Model Answer

For $\int \frac{dx}{x(x^n+1)}$, $\int \frac{dx}{x^2(x^n+1)^{(n-1)/n}}$, $\int \frac{dx}{x^n(1+x^n)^{1/n}}$: take x^n common from the bracket and put $1+x^{-n}=t$ (or t^n).

Ex: $\int \frac{dx}{x^2(x^4+1)^{3/4}} = \int \frac{dx}{x^5(1+x^{-4})^{3/4}} = -\left(1+\frac{1}{x^4}\right)^{1/4} + C.$

Grade - **Medium**

Practice map, main concept: Sheet: J-I 18-22, 24; J-II 33-36, 110; C-I 4-5; C-II 5, 14, 26 | DPP: 2(21); 5(13-14, 16); D1(21) | Question Bank: 16

Also used in: Sheet: J-II 107

Question 41

What to keep in mind while taking a power of x out as common?

Model Answer

Out of a root, $\sqrt{x^2} = |x|$ (and $\sqrt[n]{x^n} = |x|$ for even n): for $x > 0$ it is x, for $x < 0$ it is $-x$, so the answer may differ in sign on the two sides; state the interval.

Ex: $\sqrt{x^4+x^2} = |x|\sqrt{x^2+1}$, not $x\sqrt{x^2+1}$.

Grade - **Medium**

Practice map, main concept: none

Also used in: DPP: 1(17, 22) | Question Bank: 20

Miscellaneous Type

Question 42

How to integrate $\frac{ax^2+bx+c}{x^4+px^2+q}$ if $q > 0$?

Model Answer

Split into $\frac{x^2+k}{x^4+px^2+k^2}$, $\frac{x^2-k}{x^4+px^2+k^2}$ ($k=\sqrt{q}$) and $\frac{bx}{\cdots}$ (put $x^2=t$). Divide by x^2: x^2+k needs $x-\frac{k}{x}=t$; x^2-k needs $x+\frac{k}{x}=t$.

Ex: $\int \frac{x^2+1}{x^4+1}dx = \int \frac{d(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2} = \frac{1}{\sqrt{2}}\tan^{-1}\frac{x^2-1}{\sqrt{2}\,x} + C.$

Grade - **Medium**

Practice map, main concept: Sheet: J-I 28; J-II 39, 44, 100, 107, 109; J-M 13, 56; C-II 16 | Question Bank: 10

Also used in: Sheet: J-II 43; C-I 15 | DPP: 5(9) | Question Bank: 9

Integration of Irrational Algebraic Functions

Question 43

What to substitute while integrating an expression containing $(x-\alpha)(\beta-x)$, and what are dx, $x-\alpha$ and $\beta-x$ then?

Model Answer

$$x=\alpha\cos^2\theta+\beta\sin^2\theta: \quad dx=(\beta-\alpha)\sin 2\theta\,d\theta,\ x-\alpha=(\beta-\alpha)\sin^2\theta,\ \beta-x=(\beta-\alpha)\cos^2\theta$$

Used for $\sqrt{\dfrac{x-\alpha}{\beta-x}}$, $\sqrt{(x-\alpha)(\beta-x)}$ $(\beta>x>\alpha)$. Ex: $\displaystyle\int\frac{dx}{\sqrt{(x-1)(3-x)}}=2\sin^{-1}\sqrt{\frac{x-1}{2}}+C.$

Grade - Low

Practice map, main concept: Sheet: J-II 41, 77 | DPP: D1(12, 19); D4(21)

Also used in: DPP: 5(19)

Question 44

What to substitute while integrating an expression containing $(x-\alpha)(x-\beta)$, and what are dx, $x-\alpha$ and $x-\beta$ then?

Model Answer

$$x=\alpha\sec^2\theta-\beta\tan^2\theta: \quad dx=2(\alpha-\beta)\sec^2\theta\tan\theta\,d\theta,\ x-\alpha=(\alpha-\beta)\tan^2\theta,\ x-\beta=(\alpha-\beta)\sec^2\theta$$

For $\displaystyle\int\frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$ also put $x-\alpha=t^2$. Ex: $\displaystyle\int\frac{dx}{\sqrt{(x-1)(x-3)}}=2\ln\left|\sqrt{x-1}+\sqrt{x-3}\right|+C.$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 45

How to integrate $\displaystyle\int\frac{dx}{(ax+b)\sqrt{px+q}}$ and $\displaystyle\int\frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$?

Model Answer

Put $px+q=t^2$; the integral becomes a rational function of t.

Ex: $\displaystyle\int\frac{dx}{(x-3)\sqrt{x+1}}=\int\frac{2\,dt}{t^2-4}=\frac{1}{2}\ln\left|\frac{\sqrt{x+1}-2}{\sqrt{x+1}+2}\right|+C.$

Grade - Low

Practice map, main concept: DPP: 5(18-19)

Also used in: Sheet: J-II 107 | DPP: 3(19)

Question 46

How to integrate $\displaystyle\int\frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$?

Model Answer

Put $ax+b=\dfrac{1}{t}$; it reduces to $\displaystyle\int\frac{dt}{\sqrt{\text{quadratic in } t}}$ (14).

Ex: $\displaystyle\int\frac{dx}{(x+1)\sqrt{x^2+x+1}}=-\ln\left|t-\frac{1}{2}+\sqrt{t^2-t+1}\right|+C,\ t=\frac{1}{x+1}.$

Grade - Low

Practice map, main concept: Sheet: J-II 37

Also used in: Sheet: J-II 22

Question 47

How to integrate $\int \dfrac{dx}{(ax^2+b)\sqrt{px^2+q}}$?

Model Answer

Put $x = \dfrac{1}{t}$, then $p + qt^2 = y^2$ (or divide by x^3 and put $p + \dfrac{q}{x^2} = y^2$): it becomes $\int \dfrac{dy}{\text{quadratic in } y}$.

Ex: $\int \dfrac{dx}{(1+x^2)\sqrt{1-x^2}} = \dfrac{1}{\sqrt{2}} \tan^{-1} \dfrac{\sqrt{2}\,x}{\sqrt{1-x^2}} + C.$

Grade - **Medium**

Practice map, main concept: DPP: D1(20) | Question Bank: 20

Also used in: Sheet: J-II 29

Question 48

How to integrate $\dfrac{1}{(x-a)^m(x-b)^n}$ when $m+n=2$ (powers may be fractions)?

Model Answer

Take $(x-b)^2$ out: integrand $= \dfrac{1}{(x-b)^2}\left(\dfrac{x-a}{x-b}\right)^{-m}$; put $\dfrac{x-a}{x-b} = t$, $dt = \dfrac{a-b}{(x-b)^2}dx$, so it becomes $\dfrac{1}{a-b}\int t^{-m}dt$.

Ex: $\int \dfrac{dx}{(x-1)^{1/2}(x+1)^{3/2}} = \sqrt{\dfrac{x-1}{x+1}} + C.$

Grade - **Medium**

Practice map, main concept: Sheet: J-M 18; J-A 27 | Question Bank: 3

Also used in: none